

# Equivariant Systems Theory and Observer Design for Autonomous Systems

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# Outline

## 1 Kinematic mechanical systems

- Theory
- Examples
- Visualisation

## 2 Lifted Systems

- Theory
- Examples
- Matrix groups
- Visualisation
- Outputs

## 3 Equivariant Lifts

## 4 Velocity extensions

- Induced Group Actions
- Velocity Extensions

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- 4 Velocity extensions
  - Induced Group Actions
  - Velocity Extensions

# Kinematic Mechanical Systems

An  $m$ -dimensional real smooth manifold  $\mathcal{M}$  termed the *state* space.

An  $l$ -dimensional real vector space  $\mathbb{L}$  termed the *velocity* or input space.

A collection of  $n_i$ -dimensional real smooth manifolds  $\mathcal{N}_i$ , for  $i = 1, \dots, p$ , termed the *output* spaces.

A *kinematic* system is defined by state equations

$$\dot{\xi} = f(\xi, u) = f_0(\xi) + g_u(\xi), \quad \dot{\xi} \in \mathcal{M} \quad (1a)$$

$$y_i = h_i(\xi), \quad (1b)$$

For system function  $f: \mathcal{M} \times \mathbb{L} \rightarrow T\mathcal{M}$ , with

$$f(\xi, \cdot): \mathbb{L} \rightarrow T_\xi \mathcal{M}, \quad \text{affine map}$$

and smooth output maps  $h_i: \mathcal{M} \rightarrow \mathcal{N}_i$ .

## Local coordinates

In local coordinates the system map of a kinematic system has the form

$$\dot{\xi} = f(\xi, u) = f_0(\xi) + g_u(\xi) = f_0(\xi) + \sum_{i=1}^{\ell} B_i(\xi) u_i$$

for suitable smooth functions  $B_i(\xi)$ .

The elements  $u = (u_1, \dots, u_\ell)$  are associated with a basis decomposition of the vector space  $\mathbb{L}$ .

For example the unicycle model is a kinematic system

$$\frac{d}{dt} \begin{pmatrix} x \\ y \\ \theta \end{pmatrix} = \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \\ 0 \end{pmatrix} v + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} q = \begin{pmatrix} \cos(\theta) & 0 \\ \sin(\theta) & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v \\ q \end{pmatrix}$$

# Inertial Navigation System

The inertial navigation problem is posed on position, velocity and attitude.

$$\dot{p} = v$$

$$\dot{v} = g_0 e_3 + Ra$$

$$\dot{R} = R\Omega_{\times}$$

With inputs  $u = (a, \Omega, g_0)$ .

Setting  $\xi = (R, v, p)$  then

$$\dot{\xi} = (0, 0, v) + (R\Omega^{\times}, g_0 e_3 + Ra, 0) = f_0(\xi) + g_u(\xi)$$

Note that  $g_0 = 9.8$  is the gravitational constant and is treated as an input.

# Equivariance

**Definition:** A kinematic system  $\{\mathcal{M}, \mathbb{L}, \mathcal{N}_i, f(\xi, u), h_i(\xi)\}$  is said to be *equivariant* if:

There is a Lie-group  $\mathbf{G}$  with

- ① a transitive (right) group action  $\phi: \mathbf{G} \times \mathcal{M} \rightarrow \mathcal{M}$ ,
- ② a (right) group action  $\psi: \mathbf{G} \times \mathbb{L} \rightarrow \mathbb{L}$

such that for all  $X \in \mathbf{G}$  then

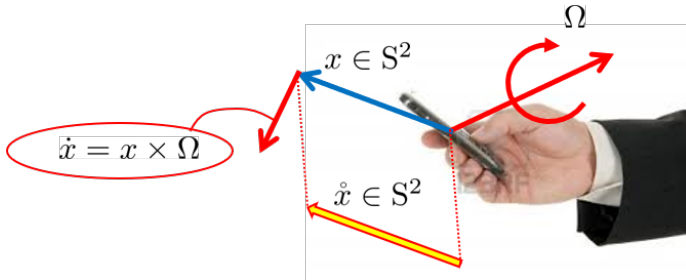
$$D\phi_X(\xi)[f(\xi, u)] = f(\phi(X, \xi), \psi(X, u)), \quad (2a)$$

It is said to have equivariant output if there are (right) group actions  $\rho^i: \mathbf{G} \times \mathcal{N}_i \rightarrow \mathcal{N}_i$  with

$$\rho^i(X, h_i(\xi)) = h_i(\phi(X, \xi)). \quad (2b)$$

The structure  $(\mathbf{G}, \phi, \psi, \rho^i)$  is termed a *symmetry*.

## Estimating direction



# State-dynamics

System kinematics on the sphere  $\xi \in \mathbb{S}^2$

$$\begin{aligned}\dot{\xi} &= \xi \times \Omega = f(\xi, \Omega) \\ y &= \xi\end{aligned}$$

i.e.  $\mathcal{N} = \mathbb{S}^2$ .

Velocity measurement (angular velocity of a rigid-body)

$$\Omega \in \mathbb{R}^3$$

Note that we have a 3-dimensional velocity with a 2-dimensional state.

# Symmetry

The group  $\mathbf{G} = \text{SO}(3)$  is a symmetry for this system with

$$\begin{aligned}\phi(Q, \xi) &= Q^\top \xi, \\ \psi(Q, \Omega) &= Q^\top \Omega, \\ \rho(Q, y) &= Q^\top y\end{aligned}$$

the associated set of right group actions.

Check the equivariance

$$\begin{aligned}D\phi_Q f(\xi, u) &= D_z|_{z=\xi} \phi_Q(z)[f(\xi, u)] = D_z|_{z=\xi} (Q^\top z)[f(\xi, u)] \\ &= Q^\top (\xi \times \Omega) = Q^\top \xi \times Q^\top \Omega \\ &= f(\phi_Q(\xi), \psi_Q(u))\end{aligned}$$

Check output compatibility  $y = h(\xi) = \xi$

$$\rho_Q(h(\xi)) = Q^\top h(\xi) = Q^\top \xi = \phi_Q(\xi) = h(\phi_Q(\xi))$$

# The unicycle with $SE(2)$ symmetry

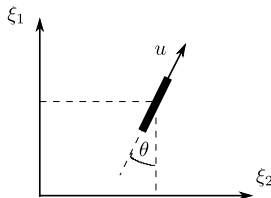
Consider the unicycle kinematics

$$\dot{x}_1 = \cos(\theta)v$$

$$\dot{x}_2 = \sin(\theta)v$$

$$\dot{\theta} = q$$

$$y = x = (x_1, x_2)$$



with state  $\xi = ((x_1, x_2), \theta) \in \mathcal{M} = \mathbb{R}^2 \times S^1$ , velocity  $u = (v, q) \in \mathbb{L} = \mathbb{R}^2$  and output  $y \in \mathcal{N} = \mathbb{R}^2$ .

Note that we have 2 dimensional velocity measurement with 3 dimensional state.

# Symmetry

$$\begin{aligned}\dot{x}_1 &= \cos(\theta)v & y &= (x_1, x_2)^\top \\ \dot{x}_2 &= \sin(\theta)v \\ \dot{\theta} &= q\end{aligned}$$

The group  $\mathbf{G} = \text{SE}(2)$  is a symmetry for this system with *right* actions

$$\begin{aligned}\phi(X, (\xi, \theta)) &= (A(\alpha)^\top \xi - A(\alpha)^\top a, \theta - \alpha) \\ \psi(X, u) &= u \\ \rho(X, y) &= A(\alpha)^\top y - A(\alpha)^\top a\end{aligned}$$

for  $u = (v, q)$  and

$$X = \begin{pmatrix} A(\alpha) & a \\ 0 & 1 \end{pmatrix} \in \text{SE}(2), \quad A(\alpha) = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}$$

# Equivariance

For  $\xi = (x_1, x_2, \theta)$  one can write

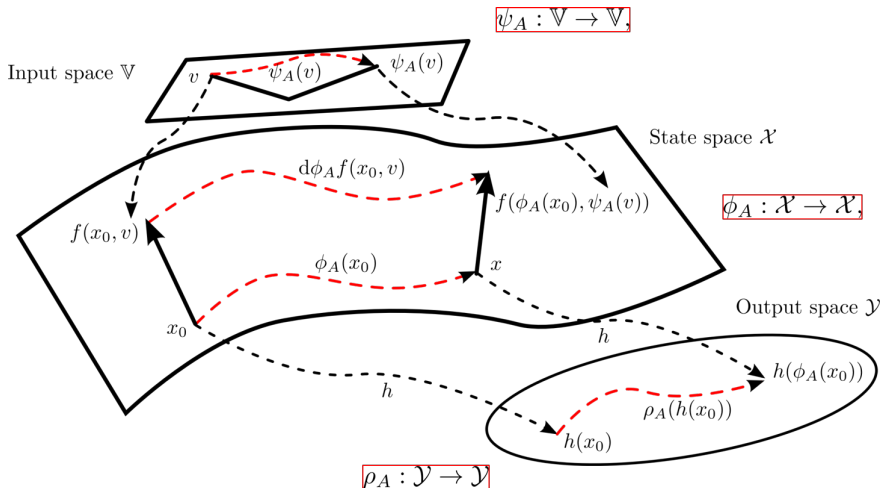
$$\dot{\xi} = \begin{pmatrix} R(\theta)\mathbf{e}_1 v \\ q \end{pmatrix} = f(\xi, (v, q))$$

Proving equivariance

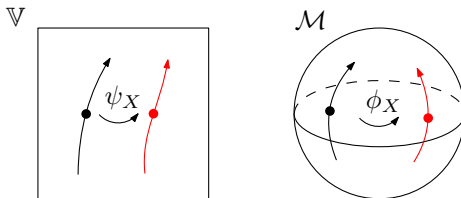
$$\begin{aligned} D\phi_X[(R(\theta)\mathbf{e}_1 v, q)] &= (A(\alpha)^\top R(\theta)\mathbf{e}_1 v, q) = (R(\theta - \alpha)\mathbf{e}_1 v, q) \\ &= \begin{pmatrix} \cos(\theta - \alpha)v \\ \sin(\theta - \alpha)v \\ q \end{pmatrix} \\ &= f(\phi_X(\xi), (v, q)) = f(\phi_X(\xi), \psi_X(v, q)) \end{aligned}$$

where the final result uses  $\psi_X(u) = \psi_X(v, q) = (v, q) = u$ .

# Algebraic visualisation of equivariance



## Behavioural visualisation of equivariance



Think of a trajectory  $u(t) \in \mathbb{L}$  for  $t \in [0, T)$  and the corresponding trajectory  $\xi_u(t) \in \mathcal{M}$  for  $\xi(0) = \dot{\xi}$ .

Then the action  $\psi_X : \mathbb{L} \rightarrow \mathbb{L}$  maps the trajectory  $u(t)$  to a new trajectory  $\psi_X(u(t))$  that corresponds to the input of a new system trajectory

$$\xi_{\psi_X(u)}(t; \phi_X(\dot{\xi})) = \phi_X(\xi(t; \dot{\xi}))$$

corresponding to the symmetry mapping  $\phi_X : \mathcal{M} \rightarrow \mathcal{M}$  applied to the old  $\xi(t)$  trajectory.

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# Lifted System

Fix the origin point  $\overset{\circ}{\xi} \in \mathcal{M}$  (Reference frame).

**Definition:** A *lifted system function* is a map

$$F_{\overset{\circ}{\xi}}: \mathbf{G} \times \mathbb{L} \rightarrow T\mathbf{G}$$

such that

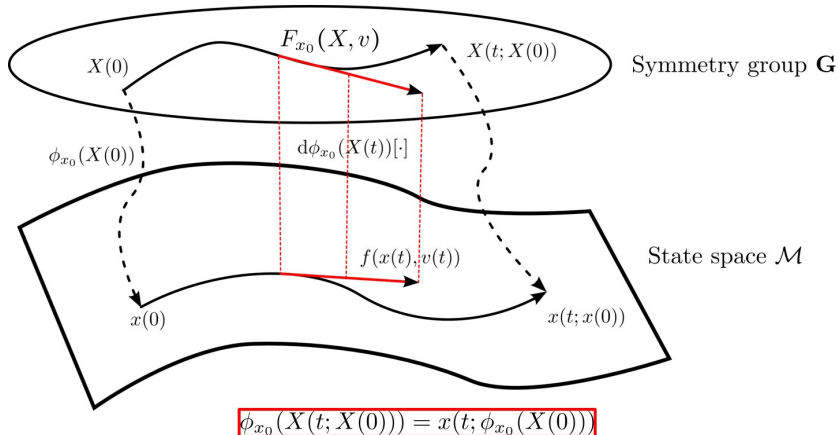
$$d\phi_{\overset{\circ}{\xi}}(X)[F_{\overset{\circ}{\xi}}(X, u)] = f(\phi_X(\overset{\circ}{\xi}), u) = f(\xi, u).$$

The lifted system is

$$\dot{X} = F_{\overset{\circ}{\xi}}(X, u), \quad X(0) = I \text{ (typically)}$$

for  $X \in \mathbf{G}$ , and  $u \in \mathbb{L}$ .

# Lifted System



# Projected dynamics

Consider the lifted kinematics on  $\mathbf{G}$  for inputs  $u(t)$ ,  $t \in [0, T]$

$$\dot{X} = F_{\xi}(X, u), \quad X(0) = I.$$

For a solution  $X(t)$ , the projected trajectory is

$$\xi(t) := \phi_{\xi}(X(t))$$

**Theorem:** The projected trajectory is the solution of the kinematic system on  $\mathcal{M}$  with initial condition  $\xi(0) = \xi^{\circ}$ .

Note that

$$\frac{d}{dt}\xi(t) = D\phi_{\xi}\dot{X} = D\phi_{\xi}(X)[F_{\xi}(X, u)] = f(\xi, u)$$

# Trivialised Lift

## Definition:

Fix an origin  $\dot{\xi}$ . A **left trivialised lift** at  $\dot{\xi}$  is a map

$$\Lambda_{\dot{\xi}} : \mathbf{G} \times \mathbb{L} \rightarrow \mathfrak{g}$$

such that

$$D\phi_{\phi_X(\dot{\xi})}(\Lambda_{\dot{\xi}}(X, u)) = f(\phi_X(\dot{\xi}), u), \quad \text{for all } \xi \in \mathcal{M}, u \in \mathbb{L}$$

A *left trivialised* **free** lift is a map

$$\Lambda : \mathcal{M} \times \mathbb{L} \rightarrow \mathfrak{g}$$

such that

$$D\phi_{\xi}(\Lambda(\xi, u)) = f(\xi, u), \quad \text{for all } \xi \in \mathcal{M}, u \in \mathbb{L}$$

A free lift  $\Lambda$  depends on the state  $\xi \in \mathcal{M}$  of the system but not on origin  $\dot{\xi} \in \mathcal{M}$ !

# A free lift defines a lifted system

Define the lifted system function

$$F_{\dot{\xi}}(X, u) := DL_X \Lambda_{\dot{\xi}}(X, u) = X \Lambda_{\dot{\xi}}(X, u) \in T_X \mathbf{G} \quad (3)$$

Then

$$D\phi_{\dot{\xi}}(X)[F_{\dot{\xi}}(X, u)] = f(\phi_X(\dot{\xi}), u) = f(\xi, u). \quad (4)$$

(compute on board)

That is

$$\dot{X} = F_{\dot{\xi}}(X, u), \quad X(0) = I$$

is a lifted system.

## Direction Estimation

The kinematics of the physical direction  $\xi \in \mathcal{M} = S^2 \subset \mathbb{R}^3$  is given by

$$\dot{\xi} = \xi \times \Omega$$

where  $\Omega \in \mathbb{L} \equiv \mathbb{R}^3$  is the angular velocity of the rigid body.

Consider the trivialised lift function

$$\Lambda(\xi, \Omega) = \begin{pmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{pmatrix} =: \Omega_{\times} \in \mathfrak{so}(3)$$

Note that  $\Lambda$  does not depend on the state  $\xi \in S^2$

# Velocity lift

Then

$$D\phi_\xi \Lambda(\xi, \Omega) = -\Omega_\times \xi = \xi \times \Omega$$

and the lifted velocity projects to the system velocity at  $\xi$  as required.  
 The lifted system function is

$$\begin{aligned} F_{\dot{\xi}}(X, \Omega) &= DL_X \Lambda(\phi_X(\dot{\xi}), \Omega) \\ &= X \Omega_\times \end{aligned}$$

The lifted system is

$$\begin{aligned} \dot{X} &= X \Omega_\times \\ y &= X^\top e_3 \end{aligned}$$

That is, the lifted system for the direction kinematics is the full attitude kinematics.

## Uncycle SE(2) example

$$\begin{aligned}\dot{x}_1 &= \cos(\theta)v & y &= (x_1, x_2)^\top \\ \dot{x}_2 &= \sin(\theta)v \\ \dot{\theta} &= q\end{aligned}$$

Using symmetry SE(2)

$$X = \begin{pmatrix} A(\alpha) & a \\ 0 & 1 \end{pmatrix} \in \text{SE}(2), \quad A(\alpha) = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}$$

$$\begin{aligned}\phi(X, (\xi, \theta)) &= (A(\alpha)^\top \xi - A(\alpha)^\top a, \theta - \alpha) \\ \psi(X, u) &= u \\ \rho(X, y) &= A(\alpha)^\top y - A(\alpha)^\top a\end{aligned}$$

# Velocity lift

We fix  $\dot{\xi} = ((0, 0), 0) \in \mathcal{M} = \mathbb{R}^2 \times S^1$  and lift  $u = (v, q) \in \mathbb{L} = \mathbb{R}^2$  by

$$\Lambda_{\dot{\xi}}(u) = \begin{pmatrix} 0 & -q & v \\ q & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \in \mathfrak{se}(2)$$

Then

$$D\phi_{\dot{\xi}}\Lambda_{\dot{\xi}}(u) = ((v, 0)^\top, q) = ((\cos(0)v, \sin(0)v)^\top, q) = f(0, u)$$

as required.

## Lifted system

The system function is now

$$\begin{aligned}
 F_{\xi}(X, u) &= DL_X \Lambda_{\xi}(\psi(X^{-1}, u)) = \Lambda_{\xi}(u)X \\
 &= \begin{pmatrix} \cos(\theta) & -\sin(\theta) & x_1 \\ \sin(\theta) & \cos(\theta) & x_2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -q & v \\ q & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} -q \sin(\theta) & -q \cos(\theta) & v \cos(\theta) \\ q \cos(\theta) & -q \sin(\theta) & v \sin(\theta) \\ 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

The lifted system is

$$\begin{aligned}
 \dot{X} &= X \Lambda_{\xi}(u) \\
 y &= X^{-1} \bar{0}
 \end{aligned}$$

## Matrix Lie group case

Let  $\mathbf{G} \subset \mathbb{R}^{n \times n}$  be a matrix group and fix an origin  $\xi \in \mathcal{M}$ .

For a matrix Lie-group

$$\Lambda_{\xi}(X, u) \in \mathfrak{g} \in \mathbb{R}^{n \times n}$$

One has for  $X \in \mathbf{G} \subset \mathbb{R}^{n \times n}$  and  $\Lambda_{\xi}(X, u) \in \mathfrak{g} \subset \mathbb{R}^{n \times n}$  then

$$DL_X \Lambda_{\xi}(X, u) = X \Lambda_{\xi}(X, u)$$

The lifted system is given by

$$\dot{X} = X \Lambda_{\xi}(X, u), \quad X(0) = I \quad (5)$$

## Example $SO(3)$ :

Recall that  $SO(3)$  acts on  $\mathcal{SO}(3)$ ,  $\phi : SO(3) \times \mathcal{SO}(3) \rightarrow \mathcal{SO}(3)$

$$\phi(R, Q) := QR \quad (\text{right action})$$

The action on the angular velocity is

$$\psi(R, \Omega) := R^\top \Omega, \quad (R^\top \Omega)^\times = R^\top \Omega^\times R$$

Define the lift to be

$$\Lambda(R, \Omega) := \Omega^\times$$

(happens not to depend on state)

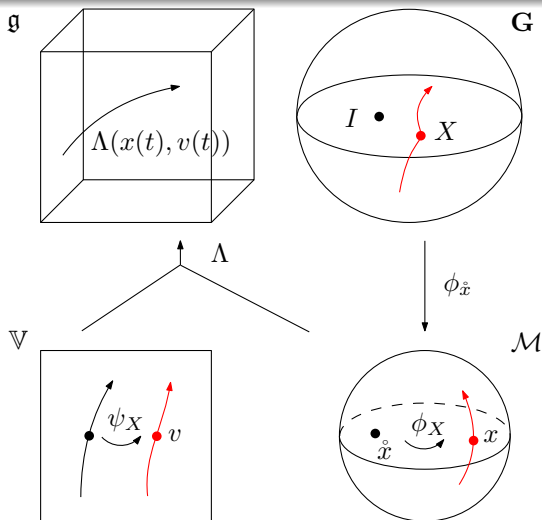
The lifted system function is

$$\begin{aligned} F_{R_0}(R, \Omega) &:= DL_R \Lambda(\phi_R(R_0), \Omega) \\ &= R \Omega^\times \end{aligned}$$

The lifted  
system is

$$\dot{R} = R \Omega^\times$$

# Behavioural interpretation of lift



$$\dot{X} = X\Lambda(\xi(t), u(t))$$

$$\Lambda(t) \in \mathfrak{g},$$

$$X(t; X(0)) \in G.$$

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$$\dot{\xi} = f(\xi(t), u(t))$$

$$u(t) \in \mathbb{L},$$

$$\xi(t; \xi(0)) \in \mathcal{M}$$

## Structure of outputs

Fix a reference  $\dot{\xi} \in \mathcal{M}$  and set  $\xi = \phi(X, \dot{\xi})$ .

For each output  $y_i$  we define a reference output  $\dot{y}_i \in \mathcal{N}_i$  by

$$\dot{y}_i := h_i(\dot{\xi}), \quad i = 1, \dots, p. \quad (6)$$

The output model can now be rewritten

$$y_i = h_i(\xi) = h_i(\phi(X, \dot{\xi})) = \rho^i(X, h_i(\dot{\xi})) = \rho^i(X, \dot{y}_i). \quad (7)$$

# Outputs

The reference  $\dot{y}_i = h_i(\dot{\xi})$  is defined using the reference  $\dot{\xi}$ , however, it is usually the reference  $\dot{y}_i \in \mathcal{N}_i$  itself that is fundamental to the observer design problem.

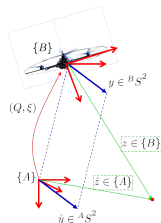
The group action  $\rho^i$  can be interpreted as encoding the relationship between two separate measurements of the same physical variable in different ‘frames’ related by the symmetry transformation.

For example in the  $SO(3)$  example

$$y_i = Q^\top \dot{y}_i$$

For fixed  $\dot{\xi}$  we define  $H_i : \mathbf{G} \rightarrow \mathcal{N}_i$  by

$$H_i(X) := \rho^i(X, h_i(\dot{\xi})) = \rho^i(X, \dot{y}_i)$$



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## Equivariant trivialised lifts

**Definition:** A trivialised lift  $\Lambda_{\xi}(X, u)$  is termed *equivariant* if

$$\Lambda_{\xi}(XZ, \psi_Z(u)) = \text{Ad}_{Z^{-1}} \Lambda_{\xi}(X, u)$$

for all  $X, Z \in \mathbf{G}$  and  $u \in \mathbb{L}$ .

For an equivariant trivialised lift one has

$$\begin{aligned} F_{\xi}^{\circ}(X, u) &= DL_X \Lambda_{\xi}^{\circ}(X, u) = DL_X \text{Ad}_{X^{-1}} \Lambda_{\xi}^{\circ}(I, \psi_{X^{-1}}(u)) \\ &= DR_X \Lambda_{\xi}^{\circ}(I, \psi_{X^{-1}}(u)) \end{aligned}$$

In matrix algebra then

$$\dot{X} = \Lambda_{\xi}^{\circ}(I, \psi_{X^{-1}}(u))X$$

That is the lift acts like a spatial velocity input to the group.

# Equivariant lifts

**Definition:** A trivialised free lift is termed *equivariant* if

$$\Lambda(\phi_X(\xi), \psi_X(u)) = \text{Ad}_{X^{-1}} \Lambda(\xi, u)$$

for all  $\xi \in \mathcal{M}$ ,  $u \in \mathbb{L}$  and  $X \in \mathbf{G}$ .

For an equivariant free lift one has

$$\begin{aligned} F_{\dot{\xi}}(X, u) &= \text{DL}_X \Lambda(\phi_X(\dot{\xi}), u) = \text{DL}_X \text{Ad}_{X^{-1}} \Lambda(\dot{\xi}, \psi_{X^{-1}}(u)) \\ &= \text{DR}_X \Lambda(\dot{\xi}, \psi_{X^{-1}}(u)) \end{aligned}$$

In matrix algebra then

$$\dot{X} = \Lambda(\dot{\xi}, \psi_{X^{-1}}(u))X$$

and the lifted system only depends on the manifold information at a single point  $\dot{\xi}$ .

## Defining an equivariant lift

Fix an origin  $\dot{\xi} \in \mathcal{M}$ .

Construct  $\Lambda_{\dot{\xi}}^0 : \mathbb{L} \rightarrow \mathfrak{g}$  such that for all  $v \in \mathbb{L}$

$$D\phi_{\dot{\xi}}\Lambda_{\dot{\xi}}^0(v) = f(\dot{\xi}, v),$$

Define

$$\Lambda_{\dot{\xi}}(X, v) := \text{Ad}_{X^{-1}} \Lambda_{\dot{\xi}}^0(\psi_{X^{-1}}(v))$$

By construction

$$\Lambda_{\dot{\xi}}(I, v) = \Lambda_{\dot{\xi}}^0(v)$$

## Finding an equivariant free lift

Fix an origin  $\dot{\xi} \in \mathcal{M}$ .

An equivariant free lift  $\Lambda(\xi, u)$  exists if and only if there exists a map  $\Lambda_{\dot{\xi}} : \mathbb{L} \rightarrow \mathfrak{g}$  such that for all  $v \in \mathbb{L}$

$$D\phi_{\xi}\Lambda_{\dot{\xi}}(v) = f(\dot{\xi}, v), \quad (8)$$

$$\text{Ad}_{S^{-1}} \Lambda_{\dot{\xi}}(v) = \Lambda_{\dot{\xi}}(\psi_S(v)), \quad \text{for all } S \in \text{stab}_{\phi}(\dot{\xi}). \quad (9)$$

This reduces the question of finding a free lift at every point  $\xi \in \mathcal{M}$  to finding one defined at a single point  $\xi$  that is invariant to action by the stabiliser.

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# System kinematics

Consider a kinematic system

$$\dot{\xi} = f(\xi, u).$$

The defining function  $f : \mathcal{M} \times \mathbb{L} \rightarrow T\mathcal{M}$  satisfies  $f(\xi, u) \in T_{\xi}\mathcal{M}$ .  
 Let  $\mathfrak{X}(\mathcal{M})$  be the set of smooth vector fields defined on  $\mathcal{M}$ . This set is  
 an (infinite dimensional) linear vector space

$$(f + g)(\xi) = f(\xi) + g(\xi), \quad (\alpha f)(\xi) = \alpha f(\xi)$$

The system function can be considered as a linear (or affine) map

$$f : \mathbb{V} \rightarrow \mathfrak{X}(\mathcal{M}), \quad u \mapsto f_u \text{ with } f_u(\xi) = f(\xi, u).$$

# The induced group action

Let  $\phi : \mathbf{G} \times \mathcal{M} \rightarrow \mathcal{M}$ , then  $\phi$  induces a group action on  $\mathfrak{X}(\mathcal{M})$ .

Define  $\phi_* : \mathbf{G} \times \mathfrak{X}(\mathcal{M}) \rightarrow \mathfrak{X}(\mathcal{M})$  by

$$\phi_*(X, f) := D\phi_X f \circ \phi_{X^{-1}}.$$

the *push forward* of  $f$ .

One has

$$\phi_*(X, f)(\xi) = D\phi_X|_{\phi_{X^{-1}}(\xi)} f(\phi_{X^{-1}}(\xi))$$

**Invariance:** A system is invariant if

$$\phi_*(X, f_u) = f_u$$

# Equivariant systems

**Equivariance:** A system is equivariant if there exists  $\psi : \mathbf{G} \times \mathbb{V} \rightarrow \mathbb{V}$  such that

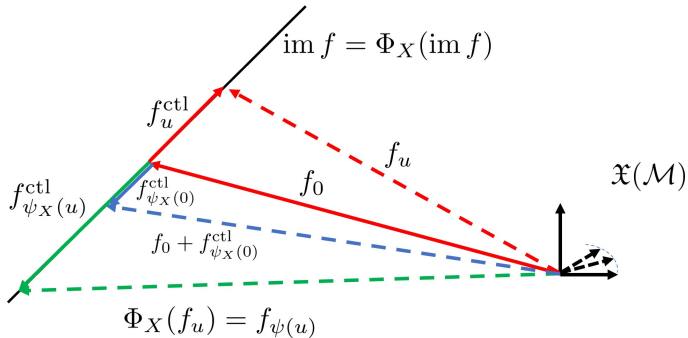
$$\phi_*(X, f_u) = f_{\psi_X(u)}$$

Note that  $\phi_*(X, f_u)$  is always defined and hence this equation fully characterises  $\psi$

$$f_{\psi_X(u)} = \phi_*(X, f_u).$$

Identifying  $f_u \leftrightarrow u$  then this is a defining relationship for  $\psi$ .

# Velocity map



# Velocity extension

The operator  $\phi_*$  is always well defined

$$\phi_*(X, f_u) \in \mathfrak{X}(\mathcal{M}).$$

however, there may be no  $w \in \mathbb{L}$  such that  $f_w = \phi_*(X, f_u)$ .

If there is no  $f_w = f_{\psi_X(u)}$  as elements in  $\mathfrak{X}(\mathcal{M})$  then one can expand the input space identifying the new input  $\psi_X(u)$  with the function

$$f_{\psi_X(u)} = \phi_*(X, f_u) = D\phi_X|_{\phi_X^{-1}(\xi)} f_u(\phi_X^{-1}(\xi))$$

Then define

$$\mathbb{V}_+ = \text{span}\{\mathbb{V}, f_{\psi_X(u)}\}$$

## Velocity extension

Based on this construction there is a new system

$$f : \mathbb{V} \oplus \text{span}\{\psi_X(u)\} \rightarrow \mathfrak{X}(\mathcal{M}).$$

The new system is equal to the old system, but now has an additional velocity input in direction  $w = \psi_X(u)$  corresponding to  $f_w = f_{\psi_X(u)} = \phi_*(X, f_u)$ .

This process is termed the **velocity extension** and can be done for any element  $X \in \mathbf{G}$  and  $u \in \mathbb{L}$ .

An alternate definition of an Equivariant system is one for which

$$\mathcal{L} = \text{span}\{\phi_*(X, f_u) \mid u \in \mathbb{L}, X \in \mathbf{G}\}$$

is finite dimensional.

## Second order kinematics

Velocity extensions are particularly important in modelling systems with integrators.

Second order linear kinematics

$$\dot{p} = v \quad (10)$$

$$\dot{v} = a \quad (11)$$

where  $a \in \mathbb{V} = \mathbb{R}$  is acceleration  $(p, v) \in \mathcal{M} \equiv \mathbb{R}^2$  is the state.

Symmetry group

$$\mathbf{G} = \mathbf{R}^3 \times \mathbf{R}^3$$

under addition.

Group action corresponds to Galilean symmetry

$$\phi_{(\alpha, \beta)}(p, v) = (p + \alpha, v + \beta).$$

## Symmetry computation

System model  $\dot{\xi} = f_u(\xi)$ ,  $\frac{d}{dt}(p, v) = (v, a) = f_a(p, v)$ .

Symmetry

$$\begin{aligned}\phi_*((\alpha, \beta), f_a(p, v)) &= D\phi_{(\alpha, \beta)} f_a \circ \phi_{(\alpha, \beta)}^{-1}(p, v) \\ &= (v - \beta, a) = (v, a) - (\beta, 0)\end{aligned}$$

Define  $\psi_{(\alpha, \beta)} a = a$  then

$$\phi_*((\alpha, \beta), f_a(p, v))(v, a) = (v, a) - (\beta, 0) = f_{\psi_{(\alpha, \beta)}(a)}(p, v) - (\beta, 0)$$

The term  $(\beta, 0)$  must be modelled as a velocity extension.

# Extended kinematics

Consider an extension of the kinematics

$$\dot{p} = v + w \quad (12a)$$

$$\dot{v} = a \quad (12b)$$

$w \in \mathbb{R}$  an extended input.

Original linear kinematics are recovered setting  $w = 0$ .

Input space is  $\mathbb{L} = \mathbb{R}^3 \times \mathbb{R}^3$  with elements  $(w, a)$ .

$$\dot{\xi} = f_{(w,a)}(p, v) = (v + w, a)$$

# Symmetry of the extended kinematics

Choose

$$\psi((\alpha, \beta), (w, a)) := (w - \beta, a).$$

Then

$$\begin{aligned}\phi_*((\alpha, \beta), f_{(w, a)}(p, v)) &= D\phi_{(\alpha, \beta)} f_{(w, a)} \circ \phi_{(\alpha, \beta)}^{-1}(p, v), \\ &= (v - \beta + w, a), \\ &= f_{(w - \beta, a)}(p, v), \\ &= f_{\psi((\alpha, \beta), (w, a))}(p, v).\end{aligned}$$

With the velocity extension included the resulting system is equivariant.

## Extension as a system embedding

Basic System has signals  $(p(t), v(t), a(t))$

$$\dot{p} = v$$

$$\dot{v} = a$$

Extended System has signals  $(p(t), v(t), a(t), w(t))$

$$\dot{p} = v + w$$

$$\dot{v} = a$$

The basic system trajectories are a subset of the possible trajectories of the extended systems.

$$\{(p(t), v(t), a(t))\} \equiv \{(p(t), v(t), a(t), w(t)) \mid w(t) = 0\}$$

If we can design an observer for the extended system then by setting  $w = 0$  we have an observer for the basic system.

# A physical point of view

## Symmetry

$$\begin{aligned}\phi((\alpha, \beta), (p, v)) &:= (p - \alpha, v - \beta) \\ \psi((\alpha, \beta), (w, a)) &:= (w - \beta, a).\end{aligned}$$

## Extended System

$$\begin{aligned}\dot{p} &= v + w \\ \dot{v} &= a\end{aligned}$$

- The velocity offset introduced by the Galilean symmetry that broke invariance is captured in the equivariance of the input through the virtual input  $w$ .
- A physical interpretation is a constant velocity offset of the reference frame as an exogenous input — for example the wind frame.